SYNOPSIS

Secure Pass – Password Security and Attack Monitoring System

Introduction

Secure Pass is a web-based security application developed using Flask (Python framework) and MySQL database. The main purpose of the system is to help users create secure passwords, analyze password strength, detect potential attacks, and monitor login activities. The application simulates modern cybersecurity concepts such as password entropy, hashing, login monitoring, and attack detection.

Objective

The main objectives of the Secure Pass project are:

- To analyse the strength of user passwords.
- To provide secure password hashing using modern algorithms.
- To detect repeated login attempts and possible attacks.
- To simulate a password leak, check similar to dark web monitoring.
- To store login and registration data securely in a database.

Technologies Used

- Programming Language: Python
- Framework: Flask
- Database: MySQL
- Database Server: XAMPP
- Password Encryption: bcrypt
- Frontend: HTML, CSS, JavaScript

System Features

1. User Registration
   Users can create an account with their name, email, and password.
2. Secure Login System
   The system verifies user credentials using encrypted passwords.
3. Password Strength Checker
   The application analyzes password entropy and categorizes it as Weak, Medium, or Strong.
4. Hash Generator
   Users can generate secure SHA-256 hashes for input text.
5. Attack Monitoring
   The system tracks login attempts and locks accounts after multiple failed tries.

6. Login Activity Logs
   All login attempts (success or failure) are recorded and can be viewed.
7. Attack Simulation
   The system simulates cyber-attacks by generating random IP addresses and tracking attempts.

Working of the System

The system allows users to register and log in through a web interface. Passwords are hashed using the bcrypt algorithm before storage. During login, the system checks credentials and records login attempts. If multiple failed attempts occur, the system detects it as a possible attack. Additionally, the application includes tools to check password strength and generate secure hashes.

Conclusion

Secure Pass demonstrates the importance of password security and attack monitoring in modern web applications. The project provides practical implementation of cybersecurity concepts such as password hashing, entropy calculation, login monitoring, and attack detection, helping users understand the importance of strong authentication systems.